

slide 8

Hinge Loss $\max(0, -y_k t_k)$

$$\mathcal{L}(w, b) = \frac{1}{N} \sum_{k=1}^N \max(0, -y_k t_k)$$

$$t_k = \begin{cases} -1 \\ 1 \end{cases} \quad y_k = w^T x_k + b$$

t_k : true label

y_k : the model's output, it's a score, not a probability, because to get the final class prediction, you still need to take $\hat{t}_k = \text{sign}(y_k)$

- for binary classification.

it punishes:

- wrong classification
- correct classification but with weak confidence

- if the model predicts correctly:

$$y_t \leq 0$$

$$\max\{0, -y_t\} \leq 0$$

$$\mathcal{L}(w, b) \leq 0$$

no penalty

else

$$y_t \leftarrow -y_t < 0$$

$$\max\{0, -y_t\} \leftarrow -y_t$$

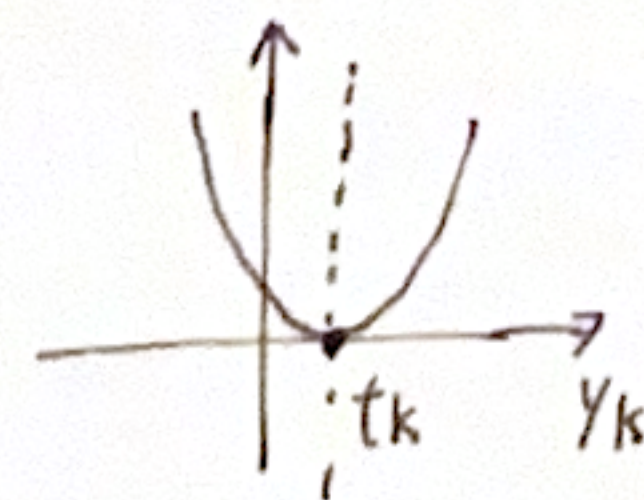
$$\mathcal{L}(w, b) \leftarrow > 0$$

penalty increases

slide 9

square loss $(y_k - t_k)^2$

$$\mathcal{L}(w, b) = \frac{1}{N} \sum_{k=1}^N (y_k - t_k)^2$$



t_k : the true target value

$y_k = w^T x_k + b$, the model's prediction

$(y_k - t_k)^2$: measures how far the prediction is from the target

slide 10

Least Square Regression

$$y = X^T w + b$$

$x \in \mathbb{R}^d$, input vector

$w \in \mathbb{R}^d$, weight vector

b : bias term

the trick here is: adding a constant dimension into the data and a corresponding weight:

$$X^T w + b = [x, 1]^T [w, b]$$

$$x' = [x, 1]^T \in \mathbb{R}^{d+1}$$

$$w' = [w, b]^T \in \mathbb{R}^{d+1}$$

$$x' w' = \begin{bmatrix} x^T \\ 1 \end{bmatrix} \cdot [w, b] = x^T w + b$$

slide 15

$$\nabla_W \mathcal{L}(W, \lambda)$$

$$= \nabla_W \left(\frac{1}{N} W^T X X^T W - \frac{2}{N} W^T X t + \lambda (\|W\|^2 - C) \right) = 0$$

$$* \|W\|^2 \leq C$$

$$\nabla_W \left(\frac{1}{N} W^T X X^T W \right) = \frac{2}{N} X X^T W$$

~~$$\nabla_W \left(-\frac{2}{N} W^T X t \right)$$~~

$$\nabla_W \left(-\frac{2}{N} W^T X t \right) = -\frac{2}{N} X t$$

$$\nabla_W (\lambda \|W\|^2) = 2 \cdot \lambda W$$

$$\frac{2}{N} X X^T W - \frac{2}{N} X t + 2\lambda W = 0$$

$$\frac{1}{N} X X^T W - \frac{1}{N} X t + \lambda W = 0$$

$$X X^T W - X t + N \lambda W = 0$$

$$(X X^T + N \cdot \lambda \cdot I) \cdot W = X t$$

$$W = (X X^T + N \lambda \cdot I)^{-1} \cdot X t$$

slide 17

feature map:

$$\Phi: \mathbb{R}^d \rightarrow \mathbb{R}^h$$

feature map is a function which takes an input vector x and outputs a new feature vector $\Phi(x)$

e.g. polynomial feature

$$\Phi(x) = (x, x^2, x^3)$$

$$\Phi(2) = (2, 4, 8)$$

slide 21

$$\text{Gaussian kernel } K(x, x_i) = \exp(-\|x - x_i\|^2 / 2\sigma^2)$$

- smooth, infinitely differentiable
- produces smooth bumps
- very flexible
- increasing flexibility = decreasing σ (scale)
- Smooth curves
- more "wiggles" when the kernel width is small

$$\text{Laplacian kernel } K(x, x_i) = \exp(-\|x - x_i\| / \sigma)$$

- less smooth than Gaussian
- sharp peaks, transitions
- produces pointy shapes
- flexible

$$\text{polynomial kernel } K(x, x_i) = (X^T X_i + C)^d$$

- Global kernel
- the entire function depends on all data
- Higher Degree $\rightarrow$ more curvature
- Degree controls complexity